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3/BSCS/A	AUTOMATA

MINIMIZATION OF DFA LAB_ACT_3

NO.1

States: {q0, q1, q2, q3, q4}

Alphabet: {0, 1}

Start State: q0

Final States: {q2, q4}

Transition Table:

State	Input 0	Input 1
=>q0	q0	q1
q1	q2	q3
(q2)	q0	q1
q3	q4	q3
(q4)	q0	q1

Minimized Transition Table

0-Equivalence

$P0 = \{\{q0, q1, q3\}, \{q2, q4\}\}$

1-Equivalence:

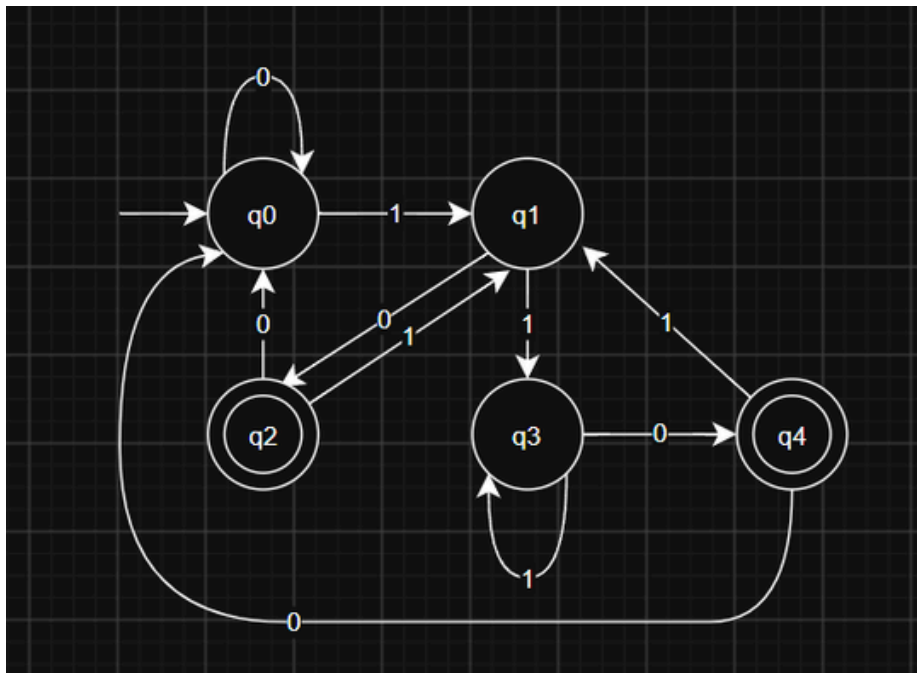
$P1 = \{\{q0\}, \{q1, q3\}, \{q2, q4\}\}$

2-Equivalence:

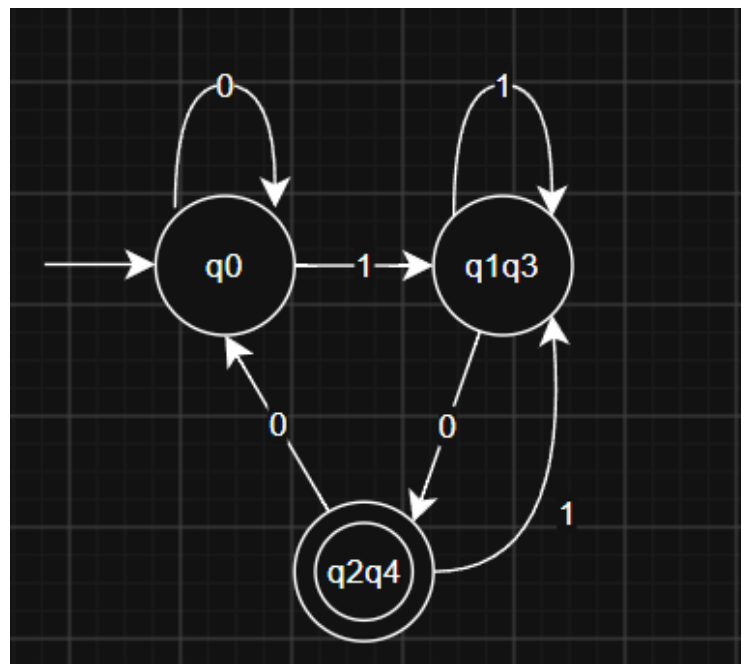
$P2 = \{\{q0\}, \{q1, q3\}, \{q2, q4\}\}$

State	Input 0	Input 1
=>q0	q0	q1 q3
q1q3	q2 q4	q1q3
(q2q4)	q0	q1 q3

BEFORE MINIMIZATION:



AFTER MINIMIZATION:



ACCEPTED:

-10
-010
-110
-0110

REJECTED:

-0
-1
-00
-101

NO.2

States: {A, B, C, D, E, E}

Alphabet: {0, 1}

Start State: A

Final States: {C,D}

Transition Table:

State	Input 0	Input 1
$\Rightarrow A$	B	C
B	A	D
(C)	E	C
(D)	E	C
E	E	E

0-Equivalence

$P_0 = \{\{A,B,E\}, \{C,D\}\}$

1-Equivalence:

$P_1 = \{\{A,B\}, \{E\}, \{C,D\}\}$

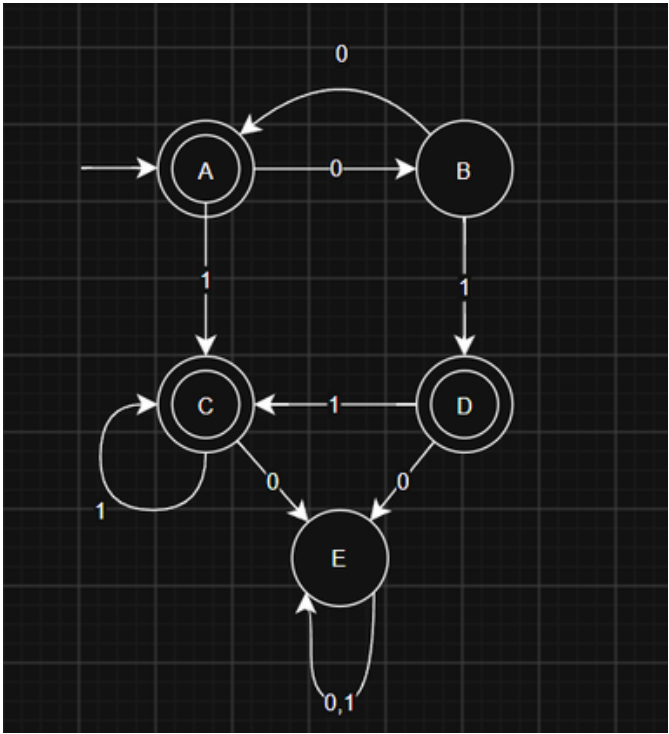
2-Equivalence:

$P_2 = \{\{A,B\}, \{E\}, \{C,D\}\}$

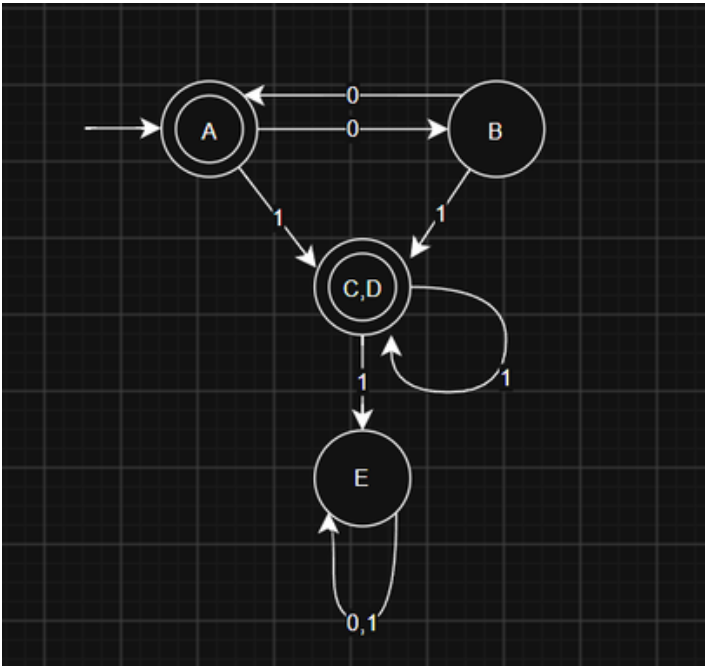
Minimized Transition Table

State	Input 0	Input 1
$\Rightarrow A$	B	(C,D)
B	A	(C,D)
(C,D)	E	(C,D)
E	E	E

BEFORE MINIMIZATION:



AFTER MINIMIZATION:



ACCEPTED:

- 1
- 01
- 001
- 11

REJECTED:

- 0
- 010
- 00
- 10